

Cymbal Copper Pty Ltd (fictional) Ridgeback Concentrator Project RB-4410	SAG Mill - Process Data Sheet Equipment tag: ML-3101	Doc No. RB-4410-PS-ML3101	Rev B
		Sheet 1 of 3	

General (cover sheet)

Parameter	Value
Service	SAG Mill
Plant area	Area 31 - SAG Milling
Equipment class	Grinding Mill
Function	Primary autogenous/semi-autogenous grinding of crushed ore; discharges via trommel/screen SC-3101.
Connected equipment	CV-3101, SC-3101, CR-3102, PP-3201A

Process design data

Parameter	Value
Diameter x EGL	12.2 x 6.7 (40 ft x 22 ft) m
Installed power	22,000 kW
Drive	Gearless motor drive (GMD), variable speed
Speed range	60 - 80 % critical
Ball charge	12 - 15 % vol
Total filling max	30 % vol
Design throughput	2,688 t/h
Specific energy	8.2 kWh/t
Liner	Cr-Mo steel shell lifters, 50 mm
Trunnion bearings	Hydrostatic / hydrodynamic, 2 x 2 pads

Revision history

Rev	Date	Description
A	2025-09-12	Issued for review
B	2026-02-20	Issued for design

Cymbal Copper Pty Ltd (fictional) Ridgeback Concentrator Project RB-4410	SAG Mill - Mechanical Data Equipment tag: ML-3101	Doc No. RB-4410-PS-ML3101	Rev B
		Sheet 2 of 3	

Mechanical design

Parameter	Value
Shell material	ASTM A516 Gr 70
Shell thickness	90 mm
Head/trunnion	Cast steel, one-piece
Feed chute	Rubber-lined, replaceable wear plates
Discharge	Trommel to SC-3101
Reline method	Mill relining machine, 2.5 t capacity
Inching drive	GMD inching mode, 0.1 rpm
Charge weight at max filling	2,450 t

Cymbal Copper Pty Ltd (fictional) Ridgeback Concentrator Project RB-4410	SAG Mill - Motor & Lubrication Data Equipment tag: ML-3101	Doc No. RB-4410-PS-ML3101	Rev B
		Sheet 3 of 3	

Drive, motor and lubrication

Parameter	Value
Motor type	Ring motor (GMD)
Rated motor power	22,000 kW
Voltage	33 kV feed / cycloconverter
Lube system	Combined hydrostatic lift / hydrodynamic lube unit
HP lift pump discharge pressure (normal)	120 bar(g)
Lube system design pressure	170 bar(g)
Bearing temperature instruments	TT-3101 / TT-3102 (RTD, duplex)
Oil grade	ISO VG 460